AI Papers Tutorial

Classic AI Papers Explained

50+ Famous AI Papers Explained in Detail

目录

第一部分

Early Machine Learning and Neural Network Foundations

第一章　Perceptron and Neural Network Origins

1.1　The Perceptron (1958)

Author: Frank Rosenblatt
Core Contribution: Proposed the Perceptron model, the雏形 of neural networks
Main Content:

- Simplest linear binary classifier
- Consists of input layer, connection weights, and output layer
- Proposed Hebb learning rule
- Proved convergence for linearly separable problems

Historical Significance: Pioneered neural network research.

1.2　Perceptrons (1969)

Author: Marvin Minsky, Seymour Papert
Core Contribution: Rigorous mathematical analysis of perceptron limitations
Main Content:

- Proved perceptrons cannot solve XOR (XOR) problems
- Strict proof of perceptron convergence theorem
- Analysis of perceptron capabilities on linearly separable problems
- Identified fundamental limitations of single-layer perceptrons

第二章　Backpropagation and Multi-layer Networks

2.1 Learning Representations by Back-propagating Errors (1986)

Author: Geoffrey Hinton, David Rumelhart, Ronald Williams
Core Contribution: Proposed Backpropagation algorithm
Main Content:

- Complete derivation of backpropagation
- Chain rule for gradient computation
- Solved effective training of multi-layer networks
- Proved multi-layer networks can learn nonlinear functions

Core Formula:

$$\delta_j^{(L)} = \frac{\partial C}{\partial a_j^{(L)}} \cdot \sigma'(z_j^{(L)})$$

2.2 Gradient-Based Learning Applied to Document Recognition (1998)

Author: Yann LeCun, Léon Bottou, Yoshua Bengio
Core Contribution: Proposed CNN for document recognition
Main Content:

- Designed LeNet-5 architecture
- Proposed convolutional and pooling layers
- Used backpropagation for training

- Achieved excellent results on MNIST handwritten digit recognition

第三章　Optimization Theory and Learning Algorithms

3.1　Adam: A Method for Stochastic Optimization (2015)

Author: Diederik P. Kingma, Jimmy Ba
Core Contribution: Proposed Adam adaptive learning rate optimizer
Main Content:

- Combines momentum and RMSProp
- Adaptive learning rate via first and second moment estimation
- Bias correction to prevent initial estimation bias

Core Formula:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$$
$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$$

3.2　Dropout: A Simple Way to Prevent Neural Networks from Overfitting (2012)

Author: Geoffrey Hinton, Nitish Srivastava, Alex Krizhevsky, Ilya Sutskever
Core Contribution: Proposed Dropout regularization
Main Content:

- Randomly drop neurons during training (with probability p)
- Equivalent to training multiple different networks and ensembling
- Effectively prevents overfitting
- Use all neurons at test time with appropriate scaling

3.3 Batch Normalization (2015)

Author: Sergey Ioffe, Christian Szegedy
Core Contribution: Proposed Batch Normalization
Main Content:

- Normalize input of each layer
- Use mini-batch statistics for mean and variance
- Introduced learnable scaling and translation parameters
- Greatly accelerated training
- Allows higher learning rates

第二部分

Deep Learning Revolution: Computer Vision Breakthroughs

第四章　ImageNet Challenge and CNN Renaissance

4.1　AlexNet (2012)

Author: Alex Krizhevsky, Ilya Sutskever, Geoffrey Hinton
Core Contribution: Won ImageNet with significant margin, sparked deep learning wave
Main Content:

- 8-layer convolutional neural network
- ReLU activation instead of Tanh/Sigmoid
- GPU parallel training
- Dropout regularization
- Data augmentation
- Reduced Top-5 error to 15.3%

4.2　VGGNet (2015)

Author: Karen Simonyan, Andrew Zisserman
Core Contribution: Demonstrated importance of network depth
Main Content:

- Proposed VGG-16 and VGG-19
- Systematic study of network depth (11-19 layers)
- Use 3×3 small convolution kernels
- Proved deeper and narrower networks work better

4.3 Inception/GoogLeNet (2015)

Author: Christian Szegedy, Wei Liu, Yangqing Jia et al.
Core Contribution: Proposed Inception module, greatly improved computational efficiency
Main Content:

- Proposed Inception module structure
- Use 1×1 convolution for dimensionality reduction
- Parallel use of different sized convolution kernels
- 22-layer deep network
- Won ImageNet 2014 classification challenge

4.4 ResNet (2016)

Author: Kaiming He, Xiangyu Zhang, Shaoqing Ren, Jian Sun
Core Contribution: Residual connections, solved training difficulty of deep networks
Main Content:

- Proposed residual learning framework
- Introduced skip connections
- Made training 152+ layer networks possible
- Won ImageNet 2015, reduced Top-5 error to 3.57%

Core Formula:

$$y = F(x, \{W_i\}) + x$$

4.5 DenseNet (2017)

Author: Gao Huang, Zhuang Liu, Laurens van der Maaten
Core Contribution: Dense connections, enhanced feature reuse
Main Content:

- Each layer directly connected to all subsequent layers
- Greatly reduced parameters

- Alleviated gradient vanishing
- Promoted feature reuse

4.6 MobileNet (2017)

Author: Andrew G. Howard, Menglong Zhu et al.
Core Contribution: Lightweight deep neural networks for mobile deployment
Main Content:

- Depthwise Separable Convolution
- Greatly reduced computation and parameters
- Two hyperparameters: width multiplier and resolution multiplier

第五章　Object Detection and Image Segmentation

5.1　R-CNN (2014)

Author: Ross Girshick, Jeff Donahue, Trevor Darrell
Core Contribution: Applied CNN to object detection
Main Content:

- Used Region Proposal method
- Extract CNN features for each proposal region
- SVM classification
- Bounding box regression
- Significantly outperformed traditional methods on PASCAL VOC

5.2　YOLO (2016)

Author: Joseph Redmon, Santosh Divvala, Ross Girshick
Core Contribution: Single-stage detector, real-time detection
Main Content:

- Object detection as regression problem
- Divided input image into $S \times S$ grid
- Each grid predicts bounding boxes and class probabilities
- End-to-end training
- Real-time speed (45 FPS)

5.3 Mask R-CNN (2017)

Author: Kaiming He, Georgia Gkioxari, Piotr Dollár, Ross Girshick
Core Contribution: Instance segmentation with added segmentation branch
Main Content:

- Added FCN segmentation head to Faster R-CNN
- Proposed RoI Align instead of RoI Pooling
- Simultaneously predict bounding boxes, classes, and segmentation masks
- Won COCO 2016 segmentation challenge

5.4 FCN (2015)

Author: Jonathan Long, Evan Shelhamer, Trevor Darrell
Core Contribution: First end-to-end pixel-level semantic segmentation
Main Content:

- Replace fully connected layers with convolutional layers
- Upsampling via deconvolution (transpose convolution)
- Skip connections to fuse different level features
- Pixel-level prediction for arbitrary size input

第三部分

Natural Language Processing and Transformer Revolution

第六章　Early RNN and NLP

6.1　LSTM (1997)

Author: Sepp Hochreiter, Jürgen Schmidhuber
Core Contribution: Long Short-Term Memory, solved long-term dependency problem
Main Content:

- Introduced gating mechanism
- Contains forget gate, input gate, output gate
- Effectively solved gradient vanishing/exploding
- Can learn long-term dependencies
- One of most important NLP architectures

Core Formula:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

6.2　Attention Mechanism (2015)

Author: Dzmitry Bahdanau, Kyunghyun Cho, Yoshua Bengio
Core Contribution: First introduced attention to seq2seq models
Main Content:

- Allows decoder to attend to different parts of source sequence
- Solved information bottleneck for long sequences
- No need to compress entire source sequence into fixed vector
- Greatly improved translation quality

Core Formula:

$$\alpha_{ti} = \frac{\exp(e_{ti})}{\sum_{j=1}^{T} \exp(e_{tj})}$$

第七章　Transformer Architecture

7.1　Transformer (2017)

Author: Ashish Vaswani, Noam Shazeer, Niki Parmar et al.
Core Contribution: Pure attention-based Transformer architecture
Main Content:

- Completely abandoned RNN and CNN, only using attention
- Multi-Head Self-Attention
- Position encoding for sequence order
- Encoder-decoder architecture
- Parallel training, greatly improved efficiency

Core Components:

- **Scaled Dot-Product Attention:** $\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$
- **Multi-Head Attention:** $\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, ..., \text{head}_h)W^O$

7.2　BERT (2018)

Author: Jacob Devlin, Ming-Wei Chang, Kenton Lee, Kristina Toutanova
Core Contribution: Bidirectional Transformer pre-training
Main Content:

- Masked Language Model (MLM) pre-training
- Next Sentence Prediction (NSP) task
- Bidirectional encoder, considers context simultaneously
- Set new records on 11 NLP tasks
- Pioneered pre-training-fine-tuning paradigm

7.3 GPT-3 (2020)

Author: Tom B. Brown, Benjamin Mann, Nick Ryder et al.
Core Contribution: Demonstrated powerful few-shot learning of large language models
Main Content:

- 175 billion parameter GPT-3 model
- Only needs few examples for downstream tasks
- No gradient updates required
- Approached or exceeded fine-tuned models on multiple benchmarks
- Demonstrated emergent capabilities

7.4 GPT-4 (2023)

Author: OpenAI
Core Contribution: Multimodal LLM, breakthrough on complex reasoning
Main Content:

- Supports image and text input
- Achieved human-level on multiple professional exams
- Significantly improved reasoning ability
- Improved safety and alignment

第八章　Large Language Models and Prompting

8.1　Chain-of-Thought Prompting (2022)

Author: Jason Wei, Xuezhi Wang, Dale Schuurmans et al.
Core Contribution: Chain-of-thought prompting technique
Main Content:

- Guide models by showing intermediate reasoning steps
- Significantly improved performance on complex reasoning tasks
- Effective on math and logical reasoning
- No additional training or gradient updates needed
- Can combine with few-shot prompting

8.2　Constitutional AI (2022)

Author: Yuntao Bai, Saurav Kadavath, Sandipan Kundu et al.
Core Contribution: Constitutional AI alignment method
Main Content:

- Use set of principles (constitution) to guide AI behavior
- AI self-criticism and improvement based on principles
- Reduced need for human labeling
- Improved model safety and harmlessness

第四部分

Reinforcement Learning

第九章　Deep Reinforcement Learning

9.1　DQN (2013)

Author: Volodymyr Mnih, Koray Kavukcuoglu, David Silver et al.
Core Contribution: First deep RL application on Atari games
Main Content:

- Combined deep CNN with Q-Learning
- Experience Replay
- Target Network for stable training
- Raw pixels as input only
- Achieved human-level on multiple Atari games

9.2　Nature DQN (2015)

Author: Volodymyr Mnih et al.
Core Contribution: Complete DQN description, human-level on 49 Atari games
Main Content:

- Complete DQN algorithm description
- Demonstrated generalization across games
- Introduced Double Q-learning to reduce Q-value overestimation

9.3 A3C (2016)

Author: Volodymyr Mnih, Adrià Puigdomènech Badia et al.
Core Contribution: Asynchronous deep RL framework
Main Content:

- Multiple agents learning in parallel
- Different agents exploring different environment variants
- Asynchronous updates to reduce non-stationarity
- Introduced Actor-Critic architecture
- Greatly improved training efficiency

第十章　AlphaGo and Chess AI

10.1　AlphaGo (2016)

Author: David Silver, Aja Huang, Chris J. Maddison et al.
Core Contribution: First defeated human professional Go player
Main Content:

- Combined deep CNN with Monte Carlo Tree Search (MCTS)
- Supervised learning to initialize policy network
- Reinforcement learning to further optimize policy network
- Trained value network to evaluate positions
- Beat Lee Sedol 4:1

Core Components:

- **Policy Network:** Predicts next move probability
- **Value Network:** Evaluates current position win rate
- **MCTS:** Combines policy and value for search

10.2　AlphaGo Zero (2017)

Author: David Silver, Julian Schrittwieser, Karen Simonyan et al.
Core Contribution: Learned completely from scratch without human data
Main Content:

- Deep neural network + MCTS
- Completely learned from self-play
- Introduced residual network architecture

- Surpassed all AlphaGo versions in 72 hours
- Defeated AlphaGo Lee 89:11 after 40 days

10.3 AlphaZero (2017)

Author: David Silver, Thomas Hubert, Julian Schrittwieser et al.
Core Contribution: Generalized AlphaGo Zero to Chess and Shogi
Main Content:

- Unified algorithm for multiple board games
- No human game data or domain-specific knowledge
- Beat Stockfish in 8 hours (Chess)
- Beat Elmo in 2 hours (Shogi)
- Demonstrated cross-domain迁移能力

第五部分

Generative Models

第十一章　Generative Adversarial Networks

11.1　GAN (2014)

Author: Ian J. Goodfellow, Jean Pouget-Abadie, Mehdi Mirza et al.
Core Contribution: GAN framework, new paradigm for generative models
Main Content:

- Adversarial training: generator vs discriminator
- Generator learns to generate realistic samples
- Discriminator learns to distinguish real from fake
- Nash equilibrium for optimality
- Amazing results on image generation

Core Formula:

$$\min_G \max_D V(D,G) = \mathbb{E}_{x\sim p_{data}(x)}[\log D(x)] + \mathbb{E}_{z\sim p_z(z)}[\log(1-D(G(z)))]$$

11.2　Conditional GAN (2014)

Author: Mehdi Mirza, Simon Osindero
Core Contribution: Conditional GAN, category-controllable generation
Main Content:

- Added condition information to generator and discriminator
- Can generate specific class samples on demand
- Extended GAN application scope
- Laid foundation for controllable generation

11.3 DCGAN (2016)

Author: Alec Radford, Luke Metz, Soumith Chintala
Core Contribution: Stable deep convolutional GAN architecture
Main Content:

- Design guidelines for training DCGAN
- Batch normalization for training stability
- Demonstrated potential for image representation learning
- Discovered vector arithmetic properties of generators

11.4 Progressive GAN (2018)

Author: Tero Karras, Timo Aila, Samuli Laine, Jaakko Lehtinen
Core Contribution: Progressive training, high-resolution image generation
Main Content:

- Start from low resolution, gradually increase network depth
- Smooth transition between stages
- First to generate 1024x1024 high-resolution faces
- Improved training stability
- Reduced mode collapse

11.5 StyleGAN (2019)

Author: Tero Karras, Samuli Laine, Timo Aila
Core Contribution: Style-based GAN generator architecture
Main Content:

- Mapping network maps latent vector to intermediate latent space $\mathcal{W}$
- Style modulation and decoupling
- Independent control of different level attributes
- Generated realistic high-resolution faces
- Convenient for image editing

第十二章　Diffusion Models

12.1　DDPM (2020)

Author: Jonathan Ho, Ajay Jain, Pieter Abbeel
Core Contribution: Diffusion probabilistic models, comparable to GANs
Main Content:

- Forward process: gradually add noise to data
- Reverse process: learn denoising to recover data
- Variational inference for lower bound optimization
- Quality comparable to GANs
- More stable training, more diverse generation

Core Process:

- **Forward (Noising):** $q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t I)$
- **Reverse (Denoising):** $p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \sigma_t^2 I)$

12.2　Diffusion Models Beat GANs (2021)

Author: Prajit Ramachandran, Tom Rainforth, Yuhta Takida
Core Contribution: Proved diffusion models surpass GANs on multiple benchmarks
Main Content:

- Proposed ADM (Adaptive Diffusion Model)
- Used larger models and CLIP guidance
- Surpassed BigGAN on FID, IS metrics
- Promoted diffusion model adoption in research community

12.3 Classifier-Free Diffusion Guidance (2022)

Author: Jonathan Ho, Curtis P. Chen

Core Contribution: Classifier-free guidance for diffusion model control

Main Content:

- Train model with both conditional and unconditional objectives
- No additional classifier needed
- Achieve guidance via mixing conditional and unconditional predictions
- Simplified conditional diffusion model training

Core Formula:

$$\tilde{\epsilon}_\theta(x_t|c) = (1+w)\epsilon_\theta(x_t|c) - w\epsilon_\theta(x_t)$$

第十三章　VAE and Other Generative Models

13.1　VAE (2014)

Author: Diederik P. Kingma, Max Welling
Core Contribution: Variational Autoencoder, trainable generative model
Main Content:

- Combined variational inference with autoencoders
- Reparameterization trick for backpropagation
- Maximize variational lower bound (ELBO)
- Continuous latent variable differentiable generative model
- Applicable to generation, interpolation, etc.

Core Formula:

$$\mathcal{L}(\theta, \phi; x) = -KL(q_\phi(z|x)||p_\theta(z)) + \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)]$$

13.2　VQ-VAE (2018)

Author: Aaron van den Oord, Oriol Vinyals, Koray Kavukcuoglu
Core Contribution: Discrete latent representation variational autoencoder
Main Content:

- Vector quantization (VQ) instead of Gaussian distribution
- Solved distribution matching problem
- Combined with autoregressive models for high-quality generation
- Discovered emergent structures in representation learning

- Foundation for subsequent discrete diffusion models

第六部分

Multimodal Learning

第十四章　Vision-Language Pre-training

14.1　CLIP (2021)

Author: Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh et al.
Core Contribution: Established connection between text and images
Main Content:

- Contrastive pre-training on 400M image-text pairs
- Predict which text matches which image
- Zero-shot transfer to downstream tasks
- Achieved zero-shot classification on multiple vision tasks
- Revealed natural language supervision potential for vision models

Core Method:

- Dual encoder architecture: image encoder + text encoder
- Contrastive learning objective
- Zero-shot inference: classification via text prompts

14.2　Flamingo (2022)

Author: Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc et al.
Core Contribution: Few-shot learning multimodal LLM
Main Content:

- Frozen LLM + trainable vision module
- Perceiver Resampler for visual features

- Supports image/video + text input
- Few-shot learning on various vision-language tasks
- Demonstrated emergent multimodal capabilities

14.3 BLIP-2 (2023)

Author: Junnan Li, Dongxu Li, Silvio Savarese, Steven Hoi
Core Contribution: Efficient vision-language pre-training bootstrapping
Main Content:

- Q-Former bridges visual encoder and LLM
- Two-stage pre-training strategy
- Frozen image encoder greatly reduced training cost
- Led on multiple VQA and image captioning tasks
- Supports various LLM as decoder

第十五章　SAM and Vision Foundations

15.1　SAM (2023)

Author: Alexander Kirillov, Eric Mintun, Nikhila Ravi et al.
Core Contribution: Foundation model for image segmentation
Main Content:

- Segment Anything task
- Supports various prompt types: points, boxes, text
- 1.1B masks, 11M image dataset
- Transformer architecture (Sam)
- Zero-shot segmentation generalization

第七部分

Optimization and Training Techniques

第十六章　Optimization Algorithm Advances

16.1　AdamW (2019)

Author: Ilya Loshchilov, Frank Hutter
Core Contribution: Correctly decoupled weight decay and adaptive learning rate
Main Content:

- Pointed out inequivalence of L2 regularization and weight decay
- Correctly implemented weight decay in Adam
- Solved Adam's generalization problem
- Became standard optimizer for training Transformers

Core Formula:

$$\theta_{t+1} = \theta_t - \eta \frac{m_t}{\sqrt{v_t} + \epsilon} - \eta \lambda \theta_t$$

16.2　RAdam (2020)

Author: Liyuan Liu, Haoming Jiang, Pengcheng He et al.
Core Contribution: Variance correction for adaptive learning rate
Main Content:

- Analyzed variance of adaptive learning rate
- Proposed variance correction for adaptive moments
- Reduced variance in early training
- No manual learning rate warmup tuning needed

16.3 SAM (2021)

Author: Pierre Foret, Ariel Kleiner, Michael Mozer, Andreas Krause
Core Contribution: Improve generalization by optimizing loss sharpness
Main Content:

- Simultaneously optimize loss value and loss sharpness
- Find flatter local minima
- Improve model generalization
- About 2x training time increase, but usually improves results

Core Formula:

$$\min_{\theta} \max_{\epsilon \in \mathcal{B}} L(\theta + \epsilon)$$

第十七章　Training Techniques

17.1　Label Smoothing (2015)

Author: Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe et al.
Core Contribution: Label smoothing regularization
Main Content:

- Convert hard labels to soft labels
- Reduce overfitting
- Improve model calibration
- Applied in Inception networks

Core Formula:

$$q_i' = (1 - \epsilon)q_i + \epsilon/K$$

17.2　MixUp (2018)

Author: Hongyi Zhang, Yann N. Dauphin, Tengyu Ma
Core Contribution: Data augmentation via linear interpolation
Main Content:

- Linear interpolation between input and label pairs
- Enhanced data diversity
- Improved robustness to adversarial examples
- Reduced memorization, improved generalization

Core Formula:

$$\tilde{x} = \lambda x_i + (1 - \lambda)x_j, \quad \tilde{y} = \lambda y_i + (1 - \lambda)y_j$$

17.3 CutMix (2019)

Author: Sangdoo Yun, Dongjin Park, Kwangyong Lee, Young Joon Lee
Core Contribution: Data augmentation via patch cutting and mixing
Main Content:

- Replace region with region from another image
- Mix labels weighted by area
- Preserve localization ability while augmenting data
- Outperformed Mixup on multiple tasks

第八部分

Transformer Evolution and New Architectures

第十八章　Vision Transformer

18.1　ViT (2021)

Author: Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov et al.
Core Contribution: Direct application of Transformer to image patch sequences
Main Content:

- Divided image into 16x16 patches
- Each patch linearly projected to vector
- Added position encoding and class token
- Used standard Transformer encoder
- Outperformed CNN with large dataset pre-training

Core Formula:

$$z_0 = [x_{cls}; x_p^1 E; x_p^2 E; ...; x_p^N E] + E_{pos}$$

18.2　Swin Transformer (2021)

Author: Ze Liu, Yutong Lin, Yue Cao, Han Hu et al.
Core Contribution: Hierarchical shifted window attention
Main Content:

- Introduced hierarchical structure for multi-scale features
- Shifted window attention mechanism
- Reduced computational complexity
- Set new records on object detection and segmentation
- Became important backbone for vision tasks

第十九章　Modern Transformer Variants

19.1　EfficientFormer (2022)

Author: Yanyu Li, Geng Yuan, Yang Wen et al.
Core Contribution: Optimized Vision Transformer latency
Main Content:

- Redesigned ViT architecture to reduce latency
- Introduced latency analysis to guide design
- Achieved efficient inference on mobile devices
- Balanced accuracy and speed

19.2　MaxViT (2022)

Author: Zhengzhong Tu, Hossein Talebi, Han Zhang et al.
Core Contribution: Combined local and global attention
Main Content:

- Multi-axis attention mechanism
- Hybrid grid attention and window attention
- Captured local and global dependencies
- Achieved SOTA on multiple tasks

第九部分

Large Language Model Architecture and Training

第二十章　LLaMA and Open Source LLMs

20.1　LLaMA (2023)

Author: Hugo Touvron, Thibaut Lavril, Gautier Izacard et al.
Core Contribution: Open-source efficient LLM
Main Content:

- 7B to 65B parameters
- Trained only on public datasets
- Optimized Transformer architecture
- Outperformed GPT-3 on multiple benchmarks
- Promoted open-source LLM development

20.2　LLaMA 2 (2023)

Author: Hugo Touvron, Louis Martin, Kevin Stone et al.
Core Contribution: Improved LLaMA and RLHF fine-tuned versions
Main Content:

- LLaMA 2: 7B to 70B parameters
- Larger training data
- Introduced Ghost Attention mechanism
- Released Llama 2-Chat dialogue model
- Supports research and commercial use

第二十一章　RLHF and Alignment

21.1　InstructGPT (2022)

Author: Long Ouyang, Jeffrey Wu, Jiangxiang Jiang et al.
Core Contribution: RLHF alignment method
Main Content:

- Collected human preference data
- Trained reward model to predict human preferences
- Used PPO algorithm to optimize language model
- Greatly improved model helpfulness and safety
- Technology behind ChatGPT

Core Steps:

- **SFT:** Supervised fine-tuning of pretrained model
- **Reward Model:** Train reward model to predict preferences
- **PPO:** Use RL to optimize generation

21.2　DPO (2023)

Author: Rafael Rafailov, Archit Sharma, Eric Mitchell et al.
Core Contribution: Simplified direct preference optimization
Main Content:

- Directly optimize preferences bypassing reward model
- Simplified training pipeline
- Reduced training instability

- Comparable or better than PPO+RL
- Widely applied in LLM fine-tuning

Core Formula:

$$\mathcal{L}_{DPO} = -\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}\left[\log\sigma\left(\beta\log\frac{\pi_\theta(y_w|x)}{\pi_{ref}(y_w|x)} - \beta\log\frac{\pi_\theta(y_l|x)}{\pi_{ref}(y_l|x)}\right)\right]$$

第十部分

Diffusion Model Evolution

第二十二章　Text-to-Image Diffusion Models

22.1　Stable Diffusion (2022)

Author: Robin Rombach, Andreas Blattmann, Dominik Lorenz et al.
Core Contribution: Latent diffusion models, greatly reduced computational requirements
Main Content:

- Diffusion in VAE latent space
- Text conditions via cross-attention
- Greatly reduced computation
- Promoted open-source image generation
- Started personal AI art era

22.2　Imagen (2023)

Author: Chitwan Saharia, William Chan, Saurabh Saxena et al.
Core Contribution: Google's high-quality text-to-image diffusion model
Main Content:

- Uses frozen T5-XXL text encoder
- Dynamic threshold processing for higher image quality
- Multi-scale resolution training
- Achieved SOTA on FID and other metrics

第二十三章　Video Generation Models

23.1　Video Diffusion Models (2022)

Author: Jonathan Ho, Tim Salimans, Alex Graves et al.
Core Contribution: Extended diffusion models to video generation
Main Content:

- 3D U-Net architecture for spatiotemporal dimensions
- Video processed as spatiotemporal volume
- Applied to unconditional and conditional video generation
- Demonstrated video prediction capabilities

23.2　Make-A-Video (2022)

Author: Sergey Tulyakov, Elena Burmistrov, Tal Meltzer et al.
Core Contribution: Video generation without text-video pairs
Main Content:

- Used pseudo-labeling to leverage text-image data
- Spatiotemporal decoder network
- Frame interpolation and super-resolution modules
- Generated high-resolution videos

第十一部分

Future Directions and Frontier Research

第二十四章　Multimodal LLMs

24.1　GPT-4V (2023)

Author: OpenAI
Core Contribution: Vision instruction following capability
Main Content:

- Vision-language multimodal model
- Understands image content and answers questions
- Supports complex visual reasoning
- Applied to document understanding, chart analysis, etc.

24.2　Kosmos-1 (2023)

Author: Shaohan Huang, Lei Li, Furu Wei, Yasheng Wang et al.
Core Contribution: Perceiving multimodal LLM
Main Content:

- Unified perception-language model
- Supports text, image, audio input
- Emergent visual capabilities
- Multimodal in-context learning

第二十五章　Continual Learning and Meta-Learning

25.1　MAML (2017)

Author: Chelsea Finn, Pieter Abbeel, Sergey Levine
Core Contribution: Model-agnostic meta-learning framework
Main Content:

- Learn good parameter initialization
- Adapt to new tasks with few gradient steps
- Applicable to classification, regression, RL
- Architecture-agnostic

Core Formula:

$$\theta^* = \theta - \alpha \nabla_\theta \mathcal{L}_{\mathcal{T}_i}(f_\theta)$$

第二十六章　Neural Architecture Search

26.1　NAS with RL (2017)

Author: Barret Zoph, Quoc V. Le
Core Contribution: Automatically search neural network architectures with RL
Main Content:

- RNN controller to generate network architectures
- Train controller to maximize validation accuracy
- Searched on CIFAR-10 and Penn Treebank
- Discovered architectures surpassing manual designs

26.2　EfficientNet (2019)

Author: Mingxing Tan, Quoc V. Le
Core Contribution: Systematic study of network scaling rules
Main Content:

- Balance depth, width, and resolution
- Proposed compound scaling coefficient
- Greatly improved efficiency
- Widely used for transfer learning

Core Formula:

$$\text{depth} : d = \alpha^{\phi}, \quad \text{width} : w = \beta^{\phi}, \quad \text{resolution} : r = \gamma^{\phi}$$

$$\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2$$

第二十七章　Summary and Outlook

AI Papers Development Timeline

1980s: Neural Network萌芽

- 1986: Backpropagation proposed, LeNet雏形
- 1997: LSTM formally published

2010s: Deep Learning Rising

- 2012: AlexNet sparked deep learning wave
- 2014: GAN pioneered new generative model paradigm
- 2015: ResNet, BatchNorm, DQN breakthroughs
- 2017: Transformer born
- 2018: BERT revolutionized NLP

2020s: Large Model Era

- 2020: GPT-3 demonstrated emergent capabilities
- 2021: ViT, DALL-E, CLIP
- 2022: Stable Diffusion, ChatGPT
- 2023: GPT-4, LLaMA, Llama 2
- 2024: Sora video generation, GPT-4o multimodal

Acknowledgments

Thank you to all researchers who advanced the field of artificial intelligence. This tutorial aims to organize and explain classic AI papers, helping readers systematically understand the development trajectory of AI.

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